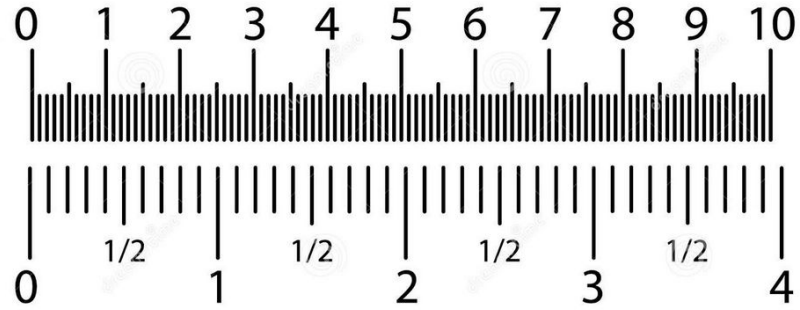


CSE200: Computer Aided Engineering Drawing

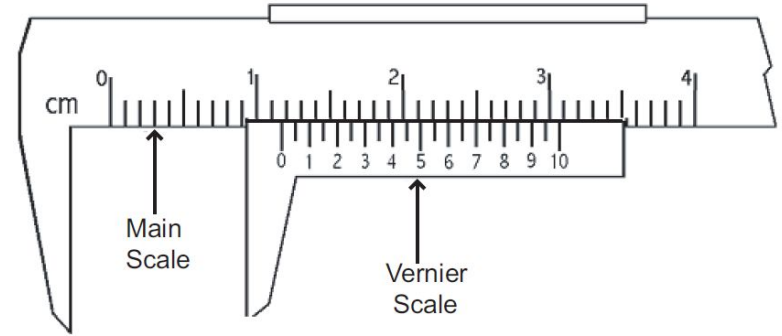
Tahsin Tariq Banna

Lecturer,
Department of Computer Science and Engineering,
East West University

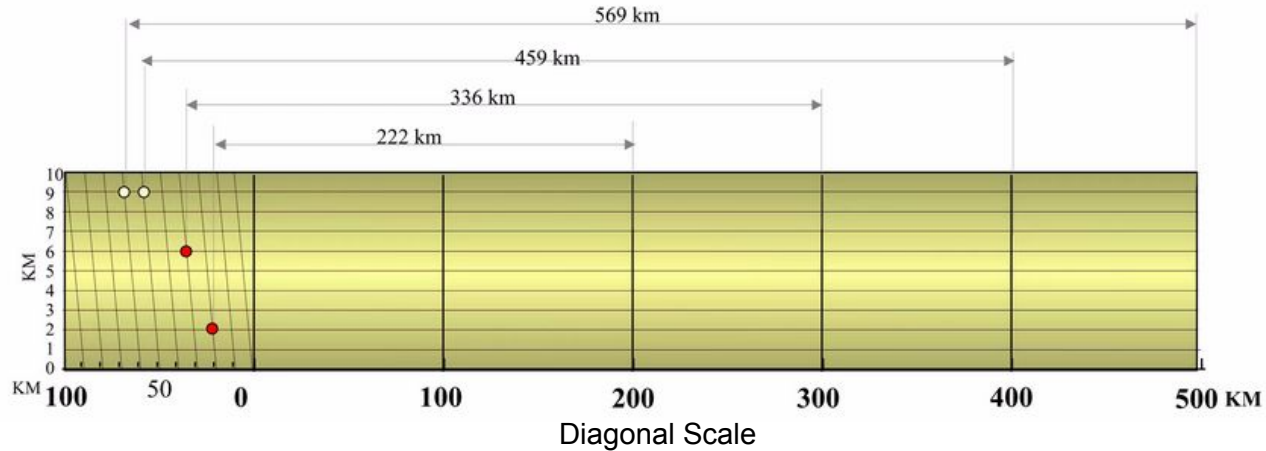
Measuring Scales



Plane Scale



Vernier Scale



Diagonal Scale

Drawing Scales

<i>Category</i>	<i>Format</i>	<i>Recommended Scales*</i>		
Enlarging Scale	$X : 1 \ (X > 1)$	50:1	20:1	10:1
		5:1	2:1	
Full Scale	$1 : 1$	1:1		
Reducing Scale	$1 : Y \ (Y > 1)$	1:2	1:5	1:10
		1:20	1:50	1:100
		1:200	1:500	1:1000
		1:2000	1:5000	1:10000

Engineering Curves

CONIC SECTIONS

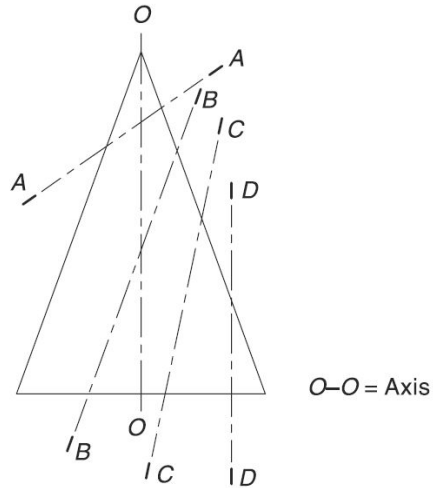


Fig. 6.1

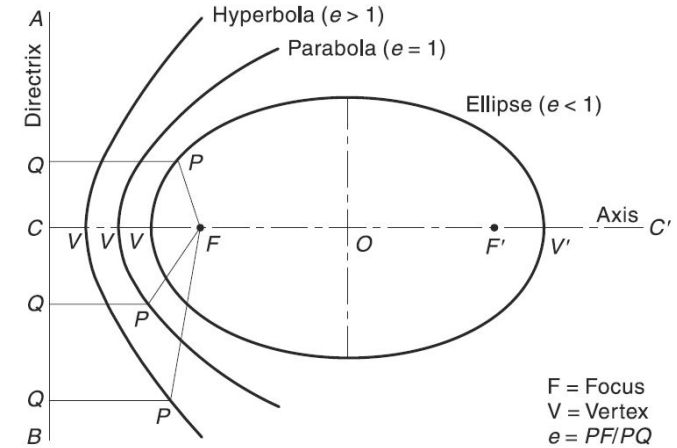
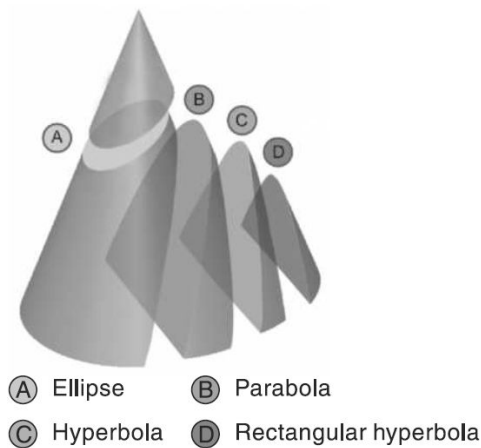


Fig. 6.2

Engineering Curves

CONIC SECTIONS



Parabola



Hyperbola

Engineering Curves

Cycloids

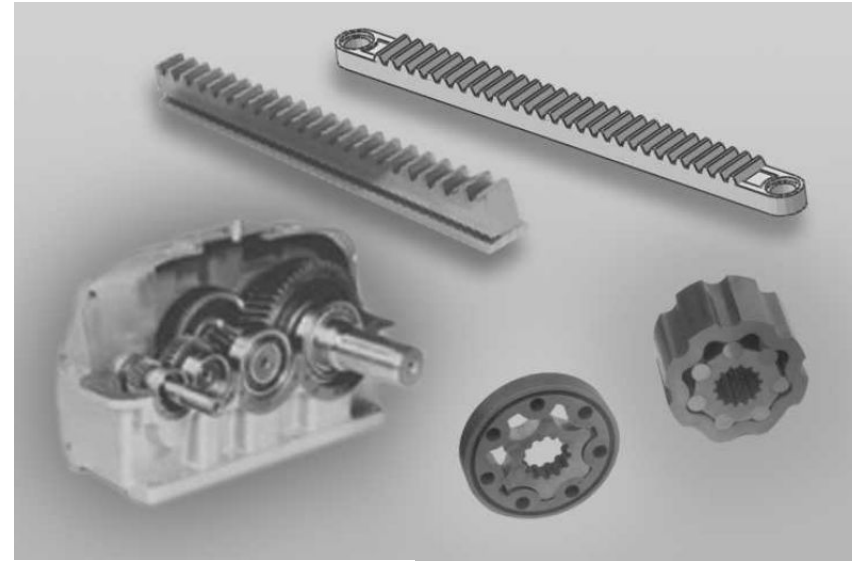
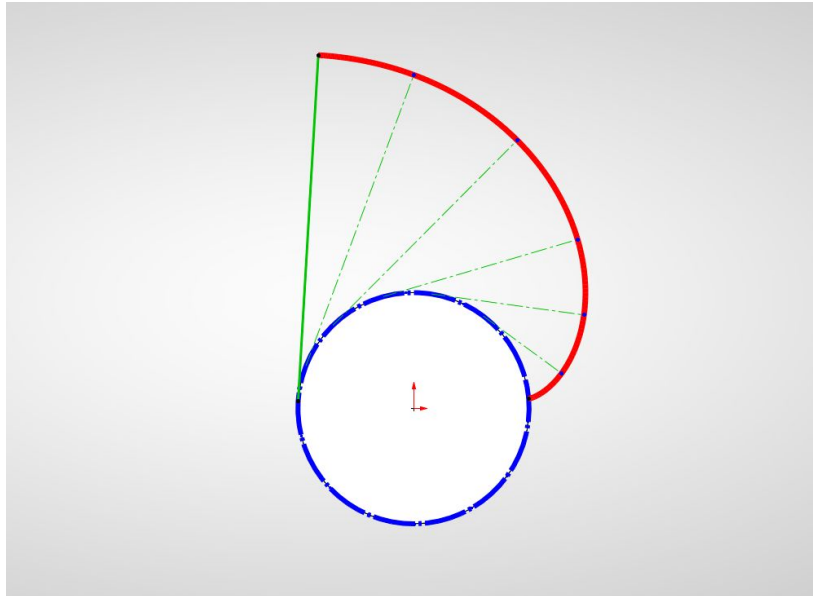
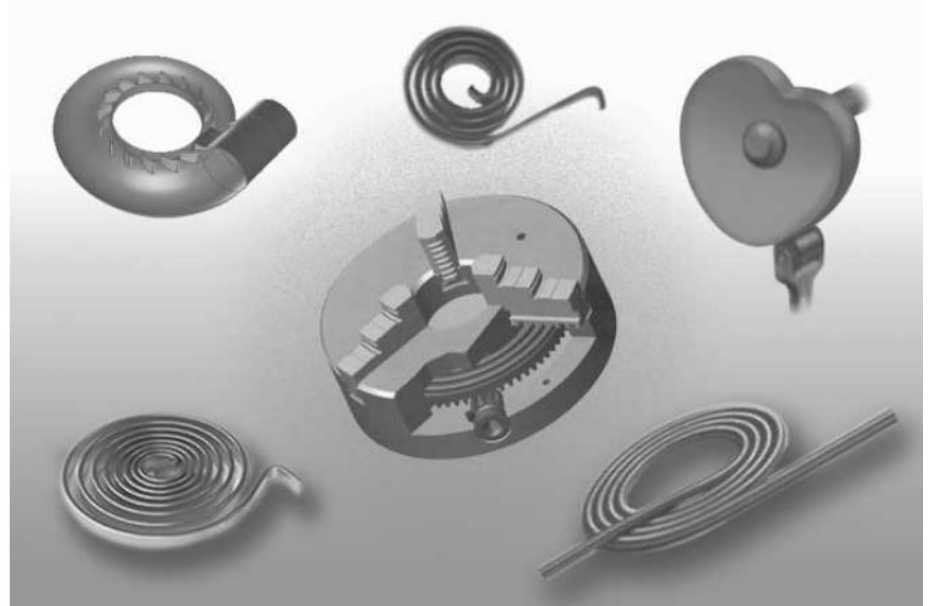


Table 6.1 Classification of Cycloidal Curve

		<i>Generating Circle</i>		
		<i>On the directing line</i>	<i>Outside the directing circle</i>	<i>Inside the directing circle</i>
<i>Generating Point</i>	<i>On the generating circle</i>	Cycloid	Epicycloid	Hypocycloid
	<i>Outside the generating circle</i>	Superior Trochoid	Superior Epitrochoid	Superior Hypotrochoid
	<i>Inside the generating circle</i>	Inferior Trochoid	Inferior Epitrochoid	Inferior Hypotrochoid

Engineering Curves

Spirals and Involutives



Engineering Curves

Helices

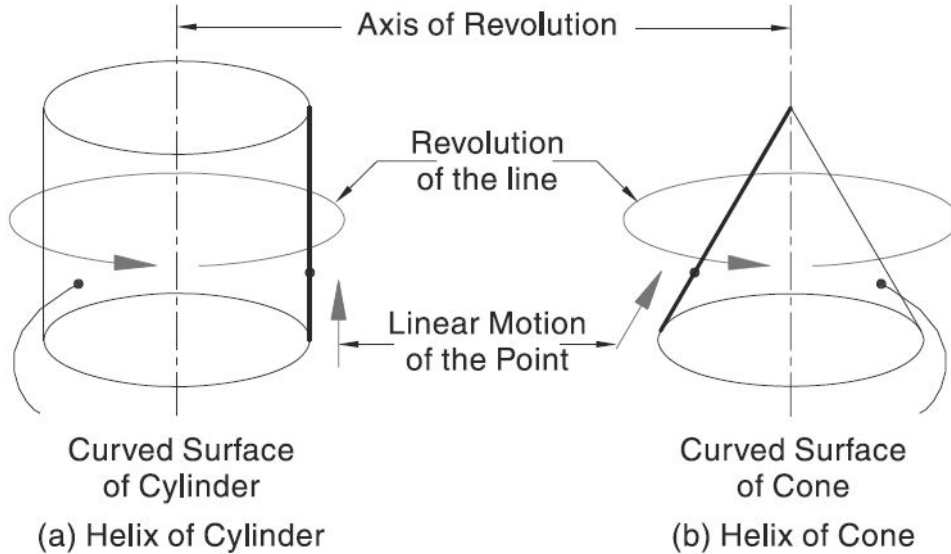
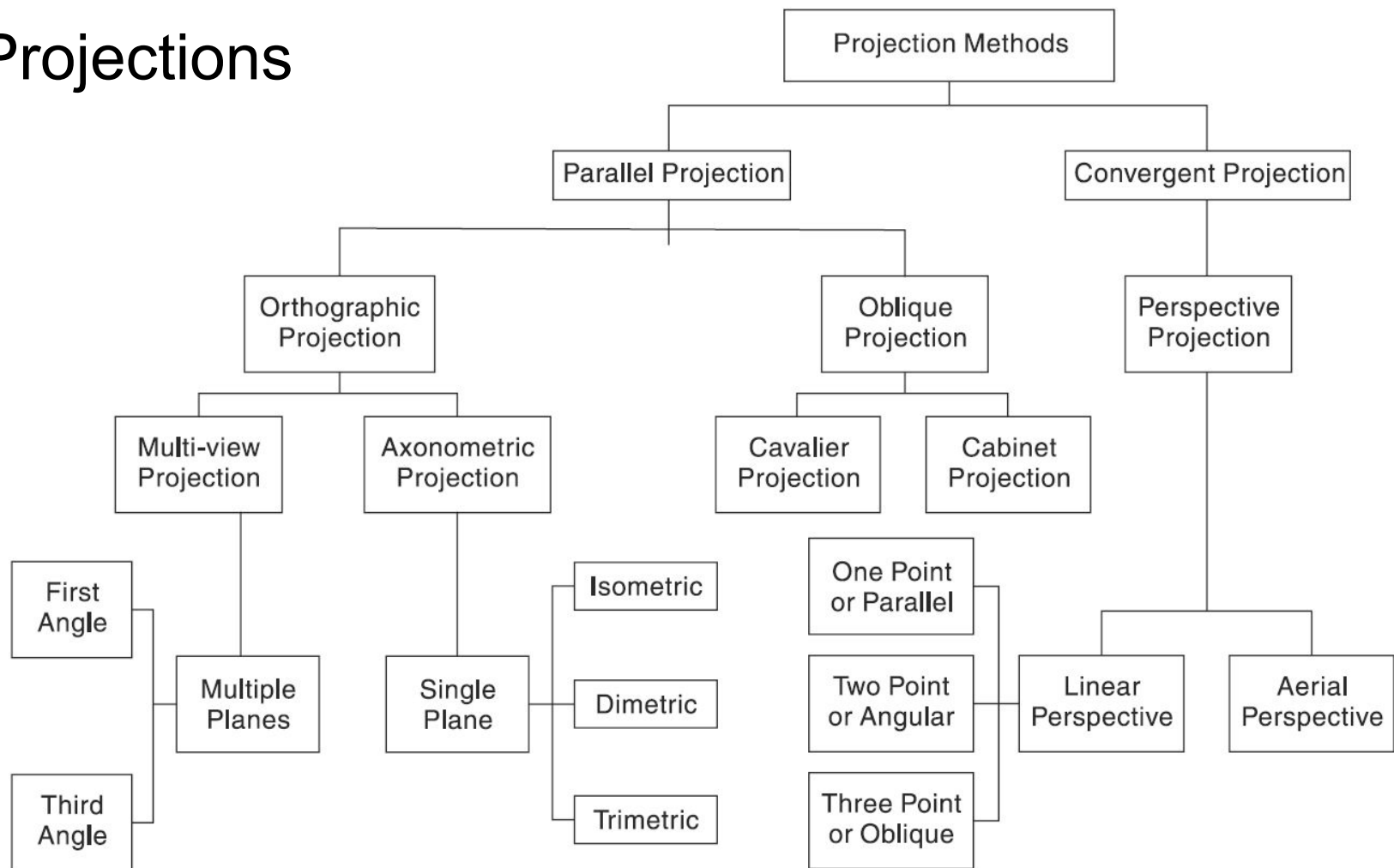


Fig. 6.34 Generation of Helix



Projections



Axonometric Projections

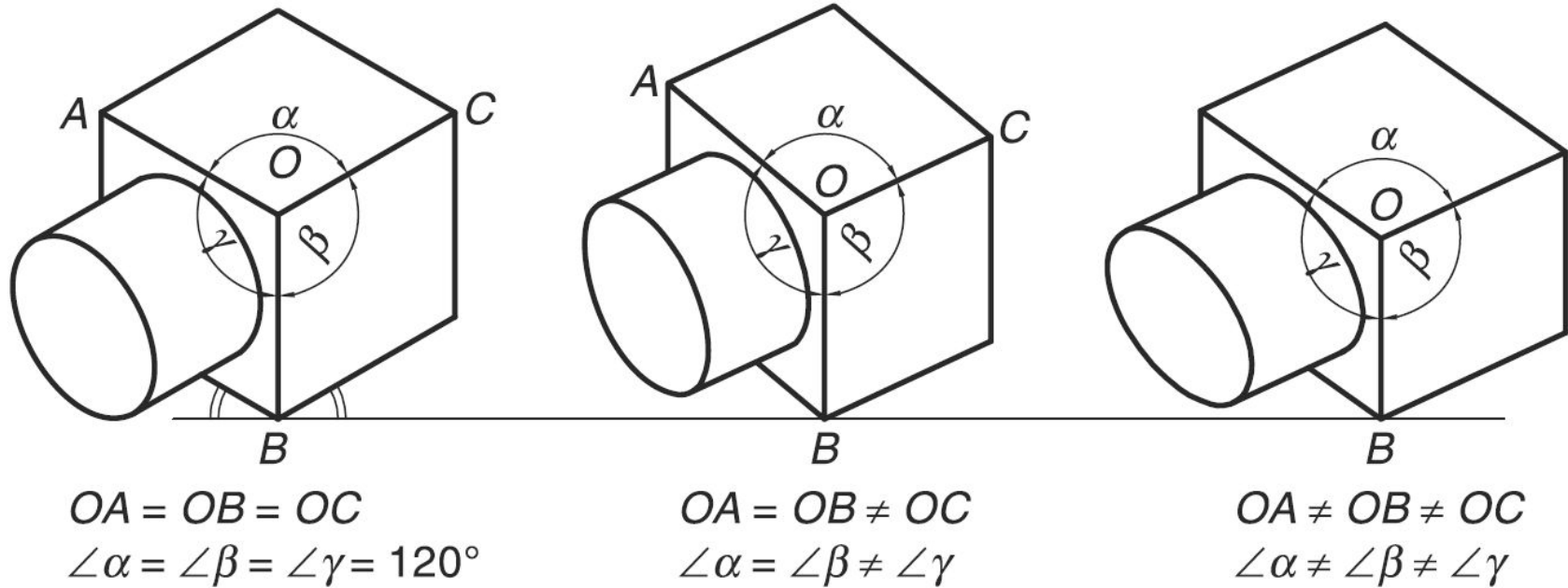


Fig. 8.4

Oblique Projections

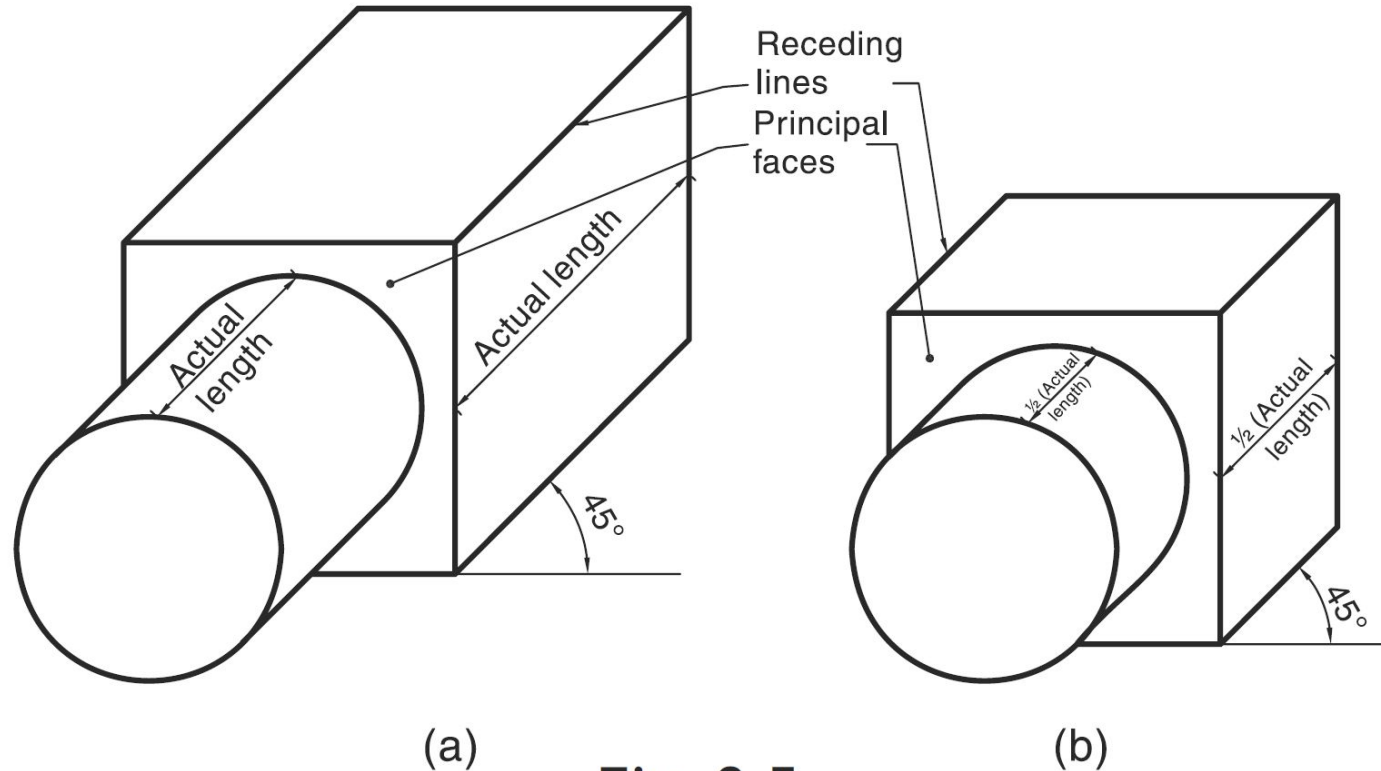
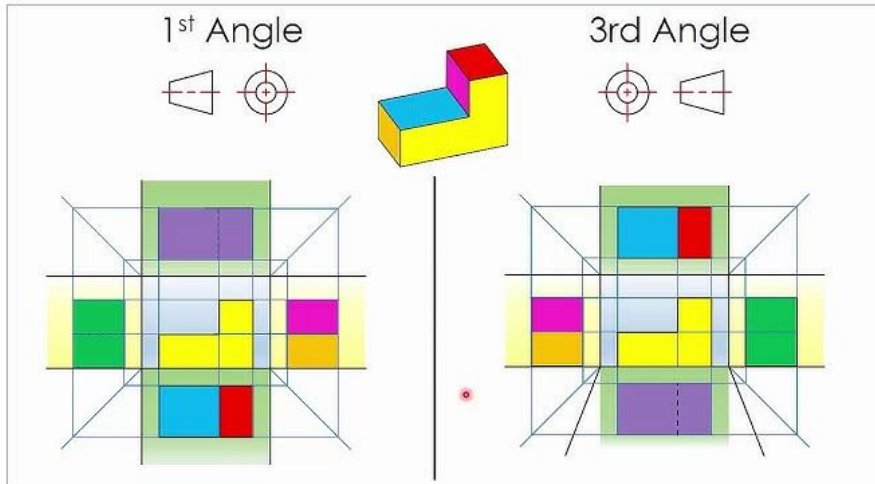
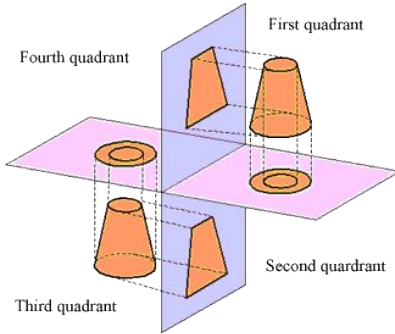
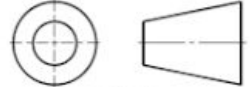
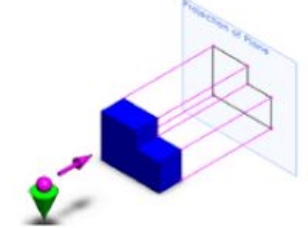
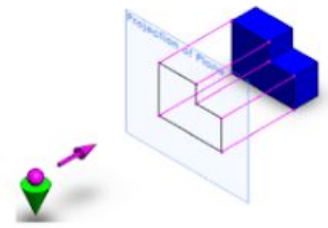
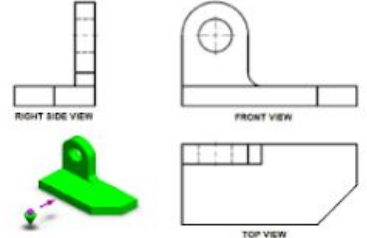
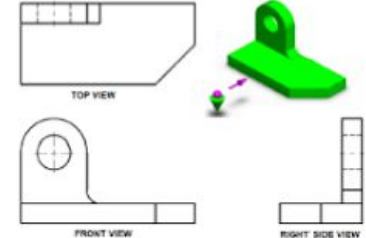


Fig. 8.5

Multiview Projections



First Angle Projection	Third Angle Projection
The object is imagined to be in first quadrant.	The object is imagined to be in third quadrant.
The object lies between the observer and plane of projection.	The plane of projection lies between the observer and object.
The plane of projection is assumed to be non transparent.	The plane of projection is assumed to be transparent.
When views are drawn in their relative position Top view comes below Front view, Right side view drawn to the left side of elevation.	When views are drawn in their relative position Top view comes above Front view, Right side view drawn to the right side of elevation.
 SYMBOL	 SYMBOL
	
 RIGHT SIDE VIEW FRONT VIEW TOP VIEW	 TOP VIEW FRONT VIEW RIGHT SIDE VIEW

Multiview Projections

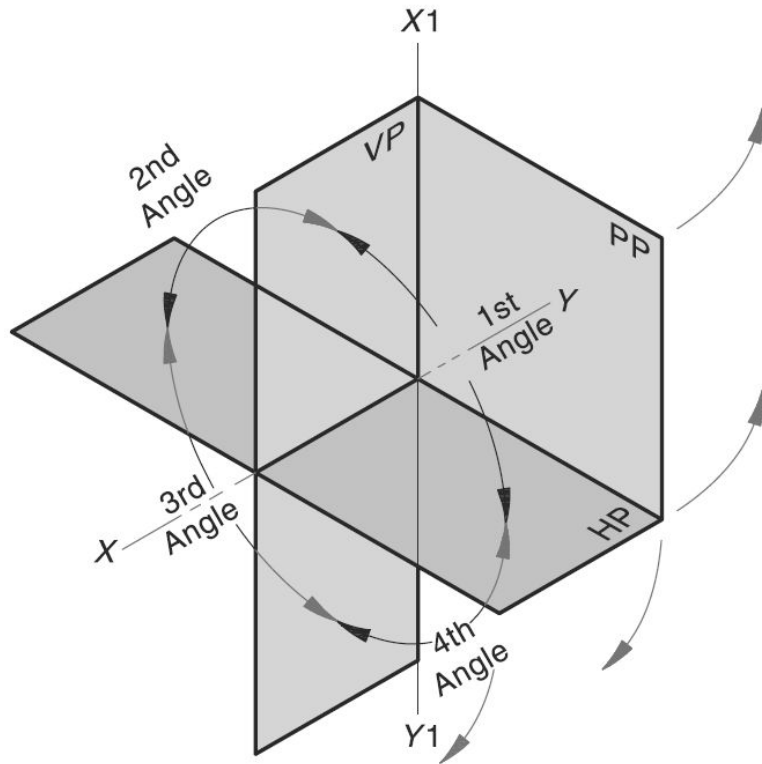
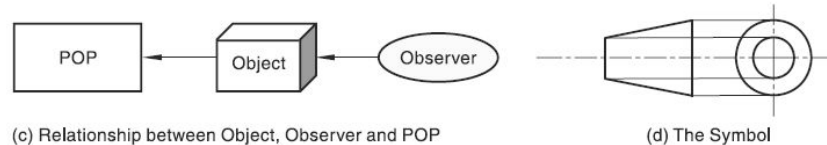
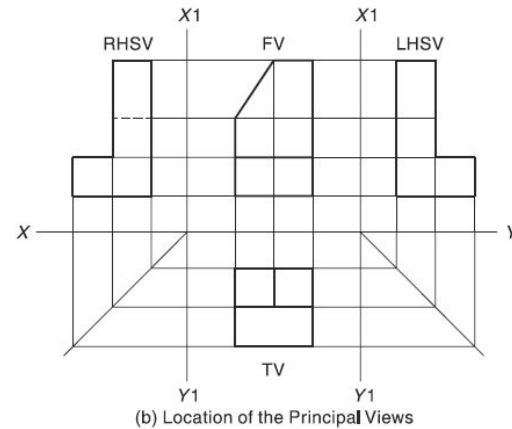
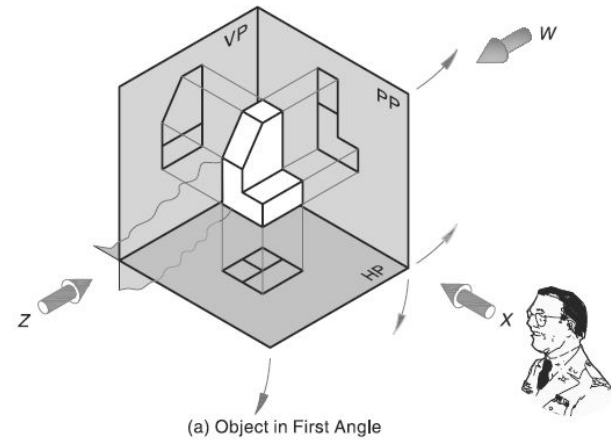
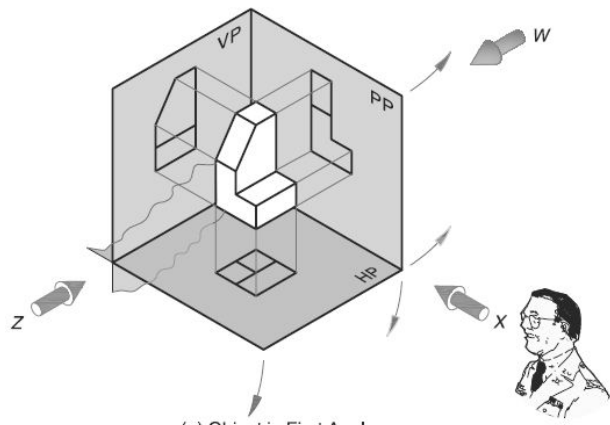
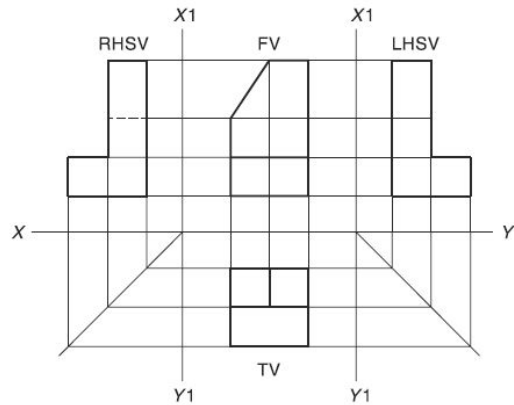


Fig. 9.1 Three principal planes

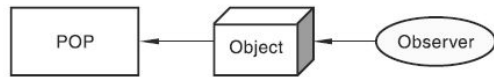




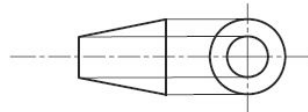
(a) Object in First Angle



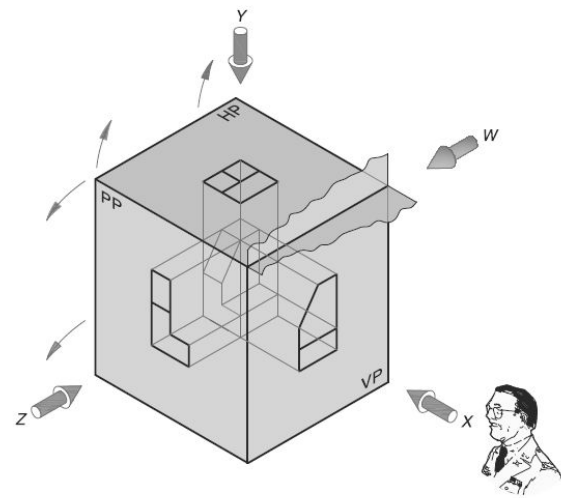
(b) Location of the Principal Views



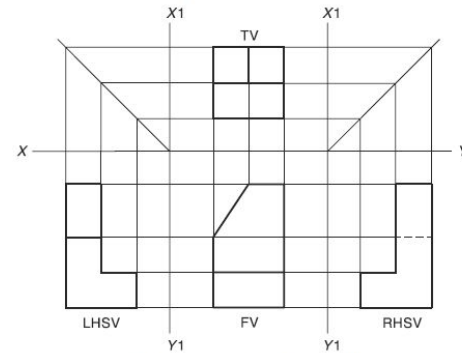
(c) Relationship between Object, Observer and POP



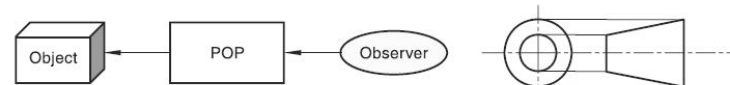
(d) The Symbol



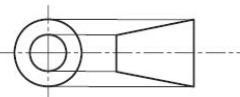
(a) Object in Third Angle



(b) Location of the Principal Views

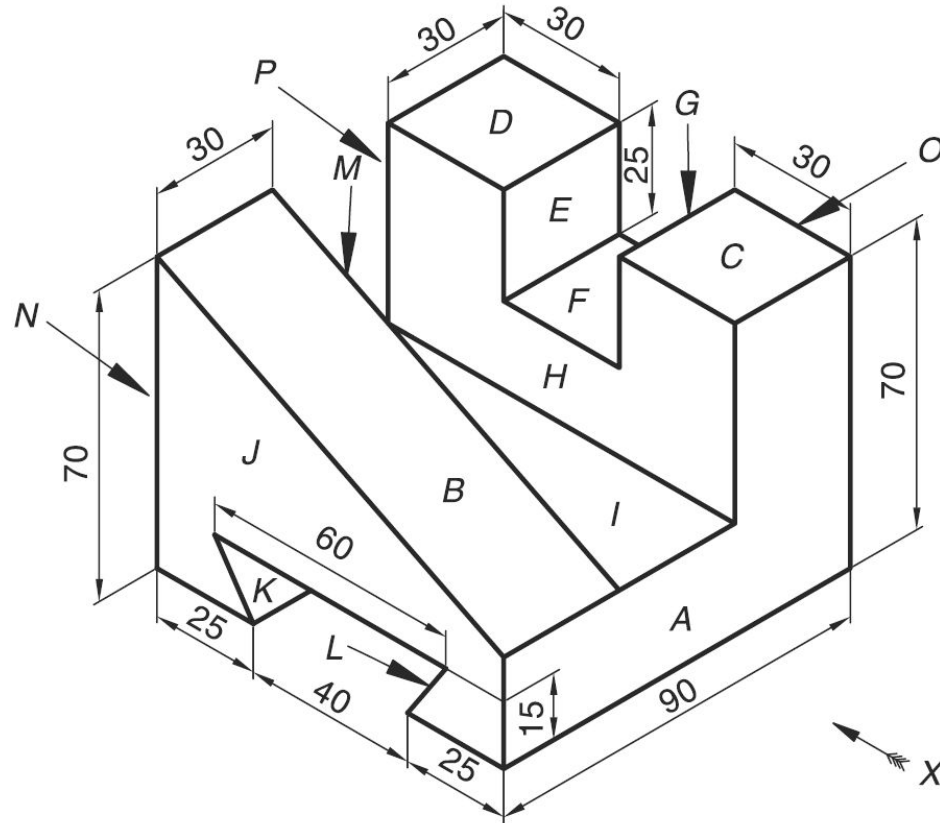


(c) Relationship between Object, Observer and POP

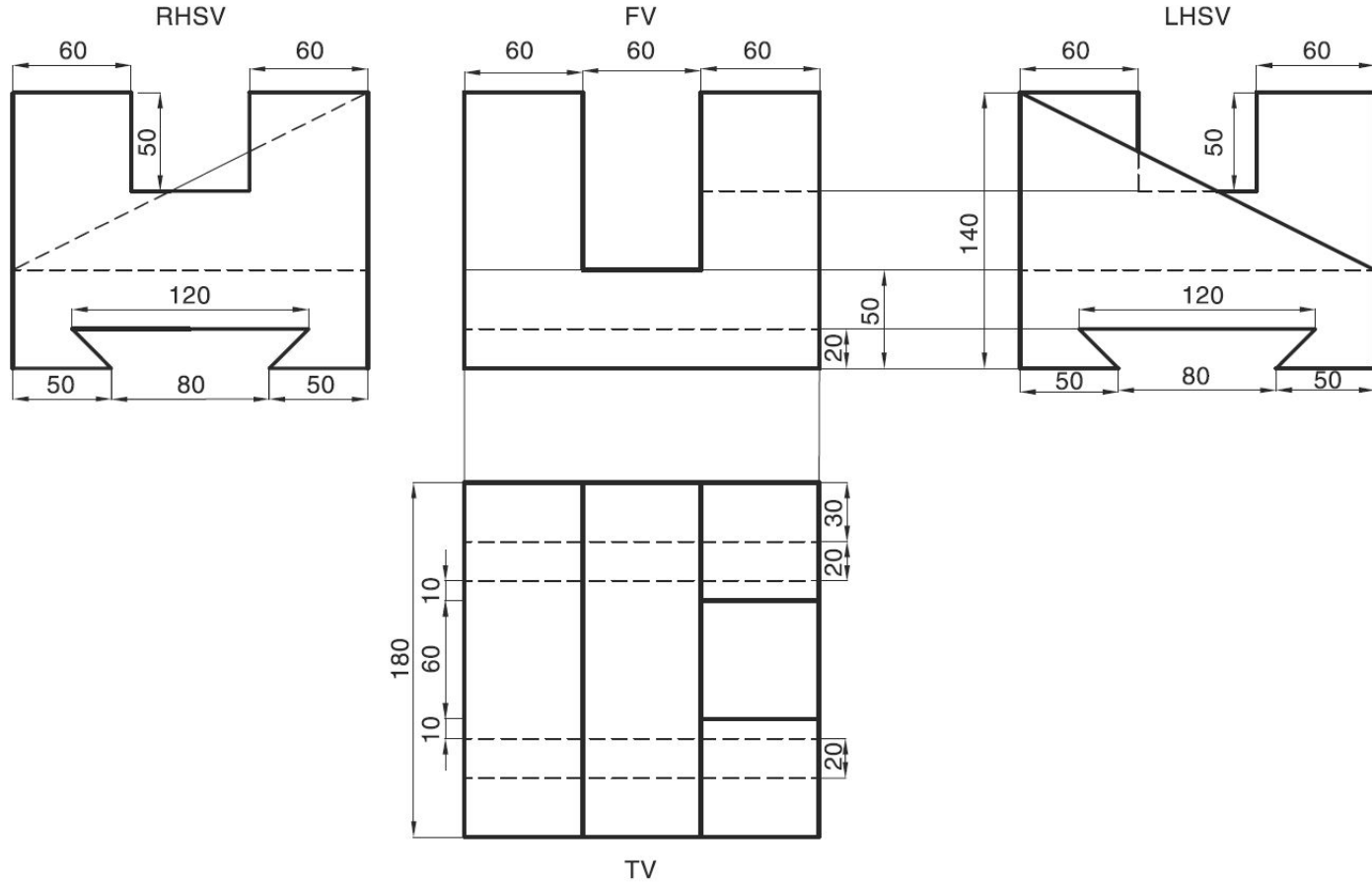


(d) The Symbol

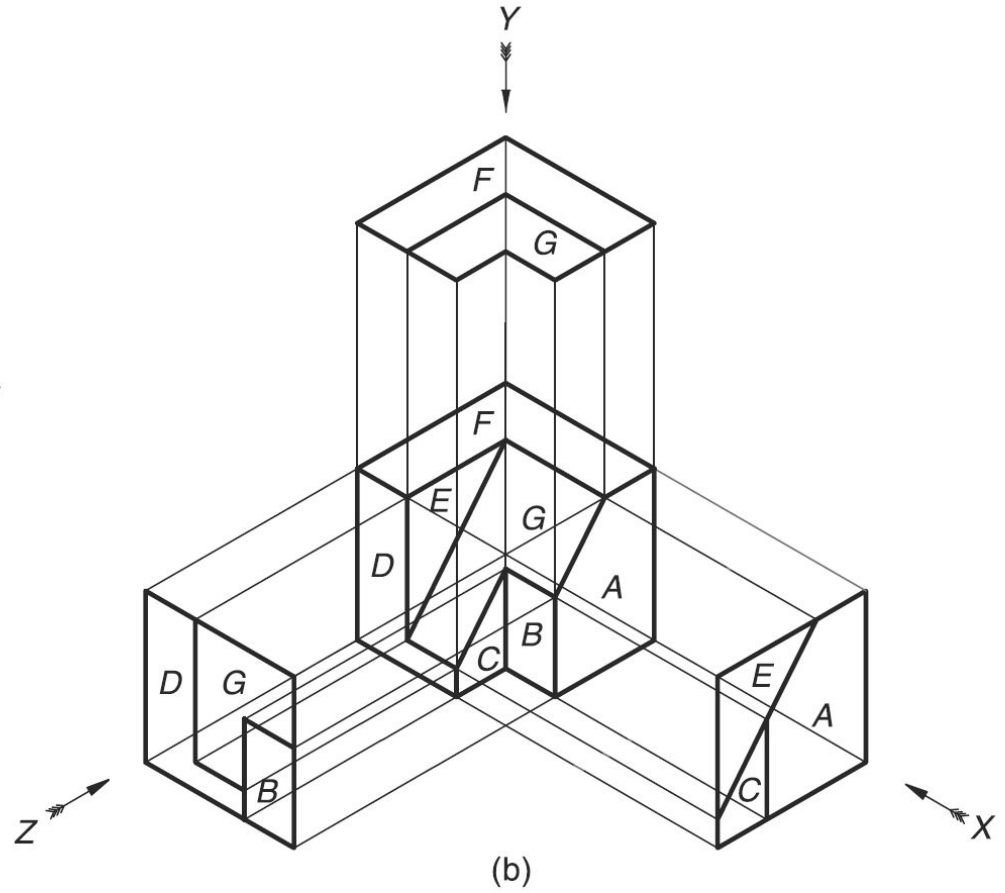
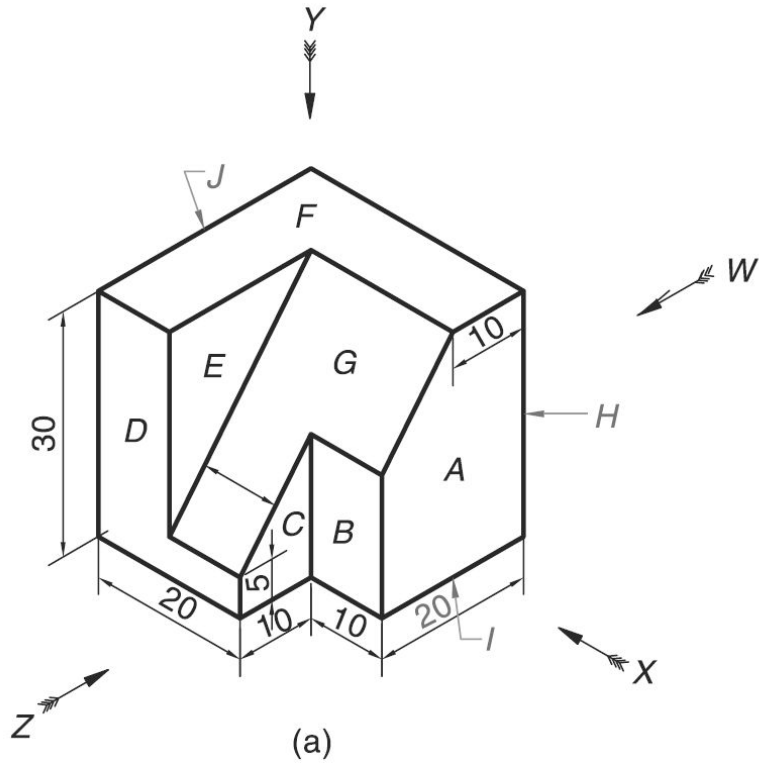
Multiview Projections



Multiview Projections

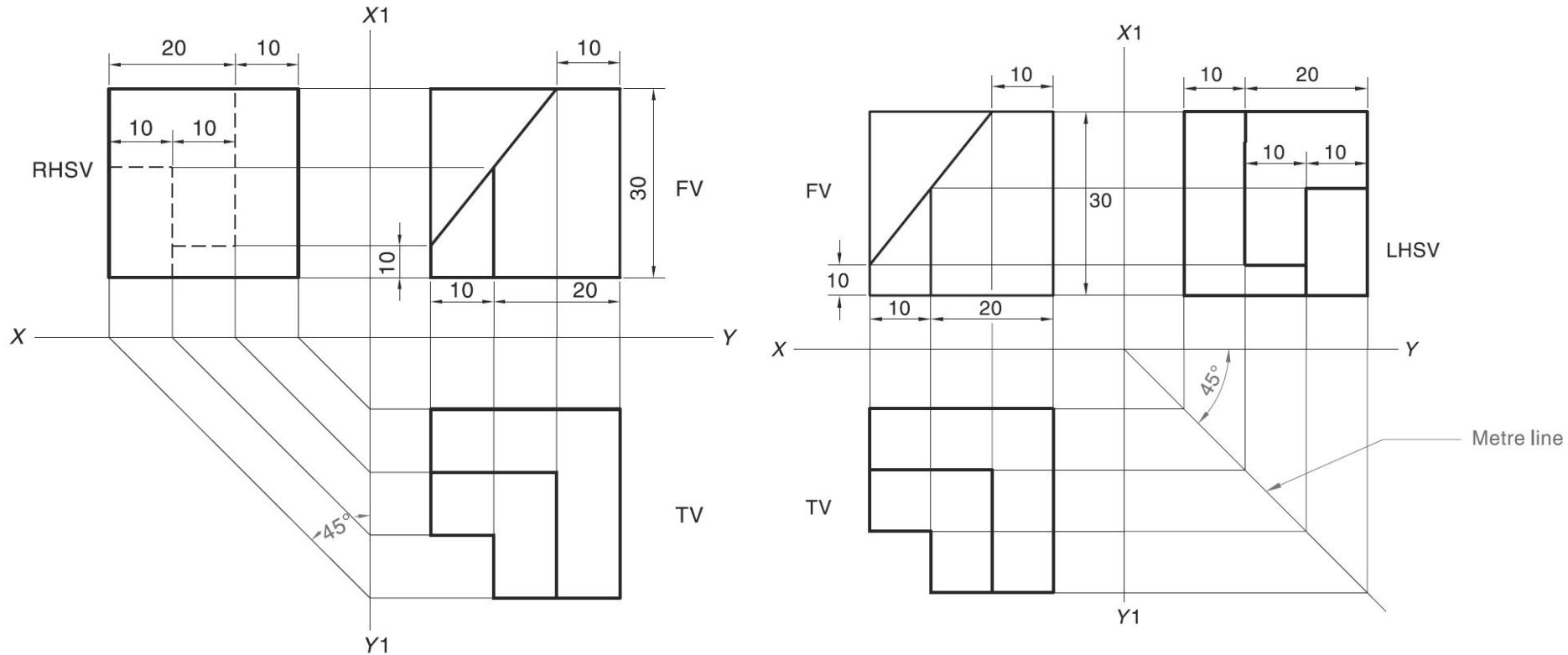


Multiview Projections



Hidden features and meter lines

Multiview Projections



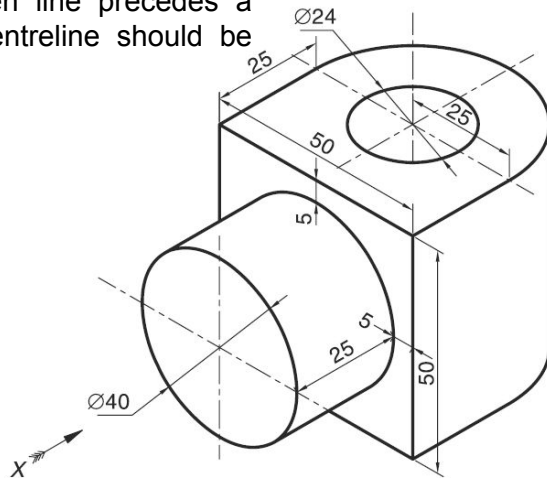
Hidden features and meter lines

Multiview Projections

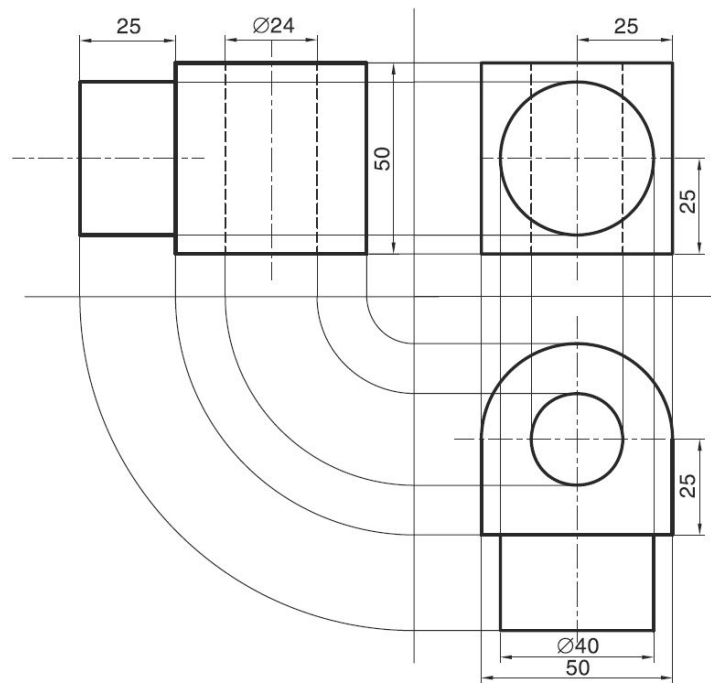
The objects having circular faces (like cylindrical projections, holes, flanges, etc.) are characterized by the centrelines of the circular features.

Wherever visible line, hidden line and centre lines overlap, the following precedence rules should be observed:

1. A visible line has precedence over a hidden line and a centreline, i.e., visible line should only be drawn if it overlaps with a hidden line and/or centreline.
2. A hidden line has precedence over a centreline, i.e., a hidden line should only be drawn if it overlaps with a centreline.
3. If a visible line or a hidden line precedes a centreline, the ends of the centreline should be drawn to show its existence.

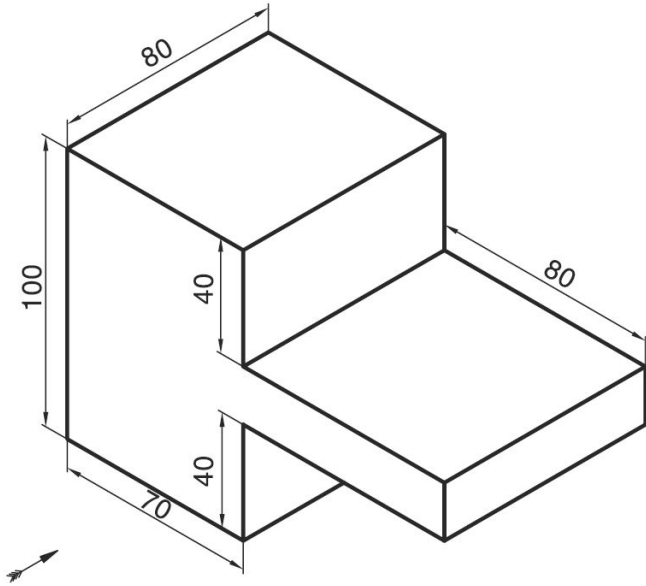


(a)

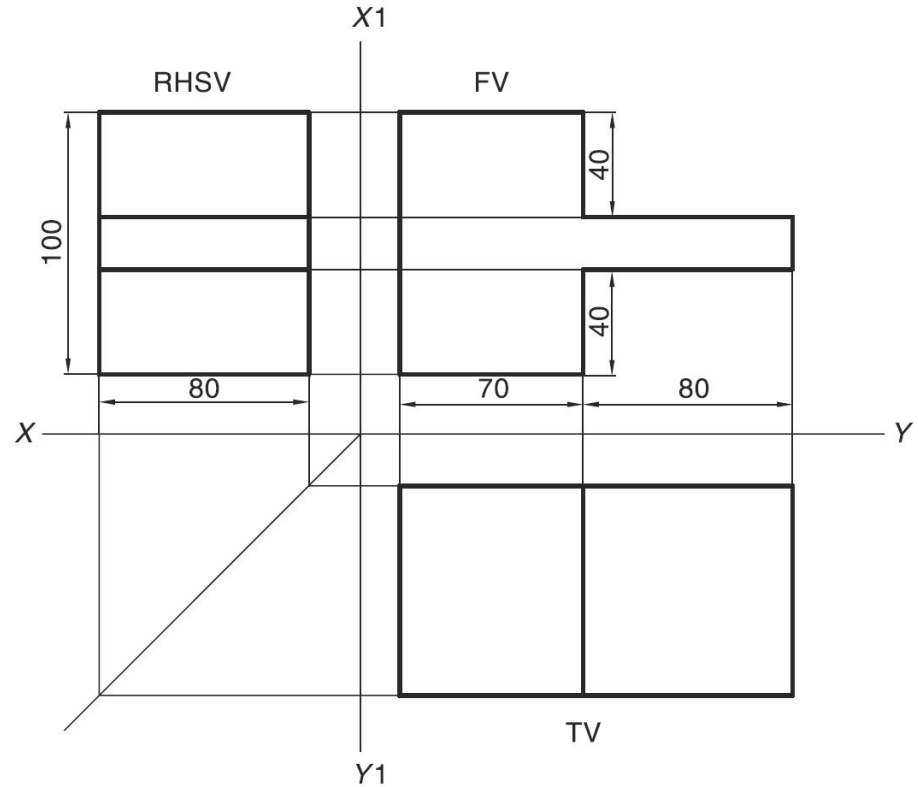


(b)

Multiview Projections

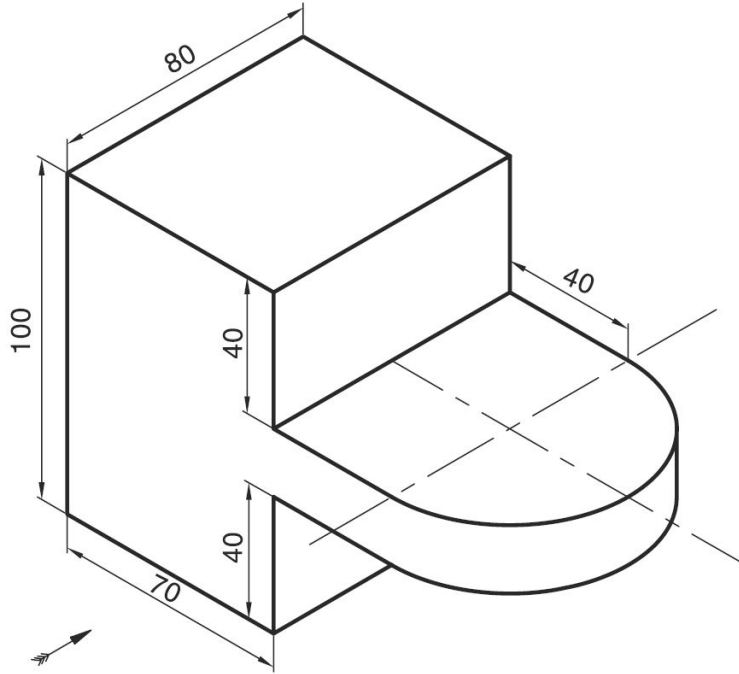


(a)

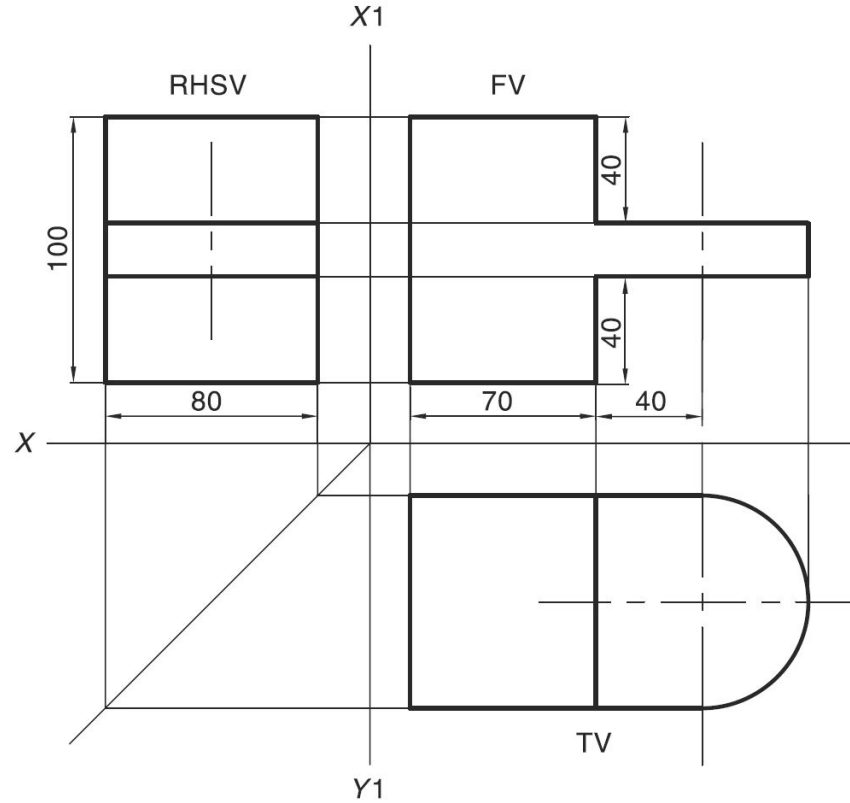


(b)

Multiview Projections



(a)



(b)